

Yiran Li

PhD, Staff Machine Learning Scientist @ Visa

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📄 <https://scholar.google.com/citations?user=T9QjVSEAAAAJ>

🏠 <https://theonlyladybug.github.io>

Staff ML Scientist specializing in foundation-model systems, LLM/VLM evaluation, and production AI applications. My work focuses on understanding and improving model behavior through prompt-driven evaluation, multimodal failure analysis, attention diagnostics, and human-centered evaluation workflows, with production experience building real-time transformer-based ML systems under strict reliability and latency constraints.

KEY SKILLS

Machine Learning: foundation models · LLM/VLM applications · multimodal learning · reinforcement learning · XGBoost

LLM/VLM Evaluation: prompt-driven evaluation · multimodal failure analysis · model behavior analysis · evaluation workflows

Model Diagnostics: attention/cross-attention analysis · SHAP-based explanation · feature attribution

Production ML & Data: real-time transformer modeling · latency-constrained ML systems · large-scale data processing

Tools: Python · PyTorch · TensorFlow · SQL · Scala · Hadoop · PySpark

PROFESSIONAL EXPERIENCE

Feb. 2025 –
Present

Visa

Staff Machine Learning Scientist

- Design and build transformer-based models for high-throughput decisioning, with emphasis on model behavior, reliability, and production constraints.
- Develop foundation-model approaches for modeling complex, evolving behavioral patterns at scale.
- Evaluate and debug production ML models through systematic error analysis, reliability checks, and performance/latency trade-off analysis.
- Explore reinforcement learning methods to improve decision consistency and robustness in real-time decision workflows.

Sep. 2018 –
Dec. 2024

University of California – Davis

Graduate Research Assistant, with Prof. Kwan-Liu Ma

- Developed LLM-assisted workflows to evaluate VLM behavior using scene graph generation, comparative analysis, and user-guided failure diagnosis.
- Built model analysis systems to study deep model behavior under perturbations using receptive field analysis and neuron-level inspection.
- Created SHAP-based tools to analyze feature importance, prediction behavior, and failure patterns in clinical ML models.
- Designed surrogate models to support interpretable analysis and optimization of domain-specific simulation outcomes.

Jun. 2023 –
Sep. 2023

Visa Research

Research Internship, with Dr. Junpeng Wang

- Built prompt-driven evaluation workflows to inspect, compare, and refine VLM-generated captions.
- Analyzed CLIP/BLIP cross-attention mechanisms using attention aggregation and gradient-weighted mapping to diagnose multimodal model behavior.
- Published findings in IEEE Transactions on Visualization and Computer Graphics (TVCG).

Jun. 2022 –
Sep. 2022

Visa Research

Research Internship, with Dr. Junpeng Wang

- Developed transformer behavior analysis methods for multi-head self-attention, including head perturbation, layer-wise pattern summarization, and attention distribution analysis.
- Identified important attention heads and layer-wise behavior patterns to support transformer model evaluation.
- Published in IEEE Transactions on Visualization and Computer Graphics (TVCG) and received the Best Paper Honorable Mention Award at the IEEE Pacific Visualization Conference.

EDUCATION

Sep. 2018 –
Dec. 2024

PhD in Computer Science

University of California – Davis, United States (GPA 4.0)

Advisor: Dr. Kwan-Liu Ma

Thesis: Visual Analytics Assistance to Interpreting, Analyzing and Improving Machine Learning Models

Sep. 2014 –
Jun. 2018

Bachelor of Science in Mathematics and Applied Mathematics, Bachelor of Arts in English

Zhejiang University (Chu Kochen Honors College), Hangzhou, China

Advisor: Dr. Zhiyi Tan

Jun. 2017 –
Sep. 2017

Exchange Undergraduate Researcher

University of California – Davis, United States

Advisor: Dr. Kwan-Liu Ma

Project: Uncertainty-Aware Visual Analytics of Dark Matter Simulation Data

Jun. 2016 –
Sep. 2016

Exchange Undergraduate Student

Harvard University, United States

PUBLICATIONS

- 2024 **Visual Analytics for Efficient Image Exploration and User-Guided Image Captioning**
Yiran Li, Junpeng Wang, Prince Aboagye, Chin-Chia Michael Yeh, Yan Zheng, Liang Wang, Wei Zhang, and Kwan-Liu Ma
TVCG IEEE PacificVis TVCG Journal Track, **Acceptance Rate: 11.5% (15 out of 131)**
- EHRFlow: A Visual Analytics Approach to Studying Healthcare professionals' Communication Effectiveness and Efficiency**
Hsiao-Ying Lu, Yiran Li, and Kwan-Liu Ma
CHASE IEEE/ACM CHASE 2024 Conference
- 2023 **How Does Attention Work in Vision Transformers? A Visual Analytics Attempt**
Yiran Li, Junpeng Wang, Xin Dai, Liang Wang, Chin-Chia Michael Yeh, Yan Zheng, Wei Zhang, and Kwan-Liu Ma
TVCG IEEE PacificVis Conference, **Best Paper Honorable Mention and published in TVCG**
- Visual Analytics of Neuron Vulnerability to Adversarial Attacks on Convolutional Neural Networks**
Yiran Li, Junpeng Wang, Takanori Fujiwara, and Kwan-Liu Ma
TIIS ACM Transactions on Interactive Intelligent Systems, Special Issue on Human-Centered Explainable AI
- A Study of Healthcare Team Communication Networks using Visual Analytics**
Hsiao-Ying Lu, Yiran Li, Brittany Garcia, Shin-Ping Tu, and Kwan-Liu Ma
ICMHI ACM International Conference on Medical and Health Informatics
- 2021 **A Visual Analytics System for Water Distribution System Optimization**
Yiran Li, Erin Musabandesu, Takanori Fujiwara, Frank J. Loge, and Kwan-Liu Ma
VIS IEEE Visualization Conference (Short Paper)
- ChartStory: Automated Partitioning, Layout, and Captioning of Charts into Comic-Style Narratives**
Jian Zhao, Shenyu Xu, Senthil Chandrasegaran, Chris Bryan, Fan Du, Aditi Mishra, Xin Qian, Yiran Li, and Kwan-Liu Ma
TVCG IEEE Transaction on Visualization and Computer Graphics
- 2020 **A Visual Analytics System for Multi-Model Comparison on Clinical Data Predictions**
Yiran Li, Takanori Fujiwara, Yong K. Choi, Katheryn Kim, and Kwan-Liu Ma
Visual Informatics IEEE PacificVis Conference (VisMeetsAI Workshop), published in Visual Informatics
- Comparative visual analytics for assessing medical records with sequence embedding**
Rongchen Guo, Takanori Fujiwara, Yiran Li, Kelly M. Lima, Soman Sen, Nam K. Tran, and Kwan-Liu Ma
Visual Informatics IEEE PacificVis Conference (VisMeetsAI Workshop), published in Visual Informatics
- Umbra: A Visual Analysis Approach for Defense Construction Against Inference Attacks on Sensitive Information**
Xumeng Wang, Chris Bryan, Yiran Li, Rusheng Pan, Yang Liu, Wei Chen, and Kwan-Liu Ma
TVCG IEEE Transaction on Visualization and Computer Graphics
- 2018 **Visual Analysis of Simulation Uncertainty Using Cost-Effective Sampling**
Annie Preston, Yiran Li, Franz Sauer, and Kwan-Liu Ma
LDAV IEEE Symposium on Large Data Analysis and Visualization

AWARDS AND HONORS

- 2023, 2024 Spring Research Fellowship from the Graduate Group in Computer Science of UC Davis
- 2023 Best Paper Honorable Mention on IEEE PacificVis

SERVICE AND OUTREACH

Program Committee

- 2025, 2023 Workshop on Visual Analytics in Healthcare (VAHC)

Conference and Journal Reviewer

- 2024 TVCG, Visual Informatics, IEEE PacificVis (TVCG Track), AAAI ICWSM
- 2023 IEEE VIS, IEEE PacificVis, ChinaVis, IEEE VIS VAHC Workshop